Proof of inequality (*) for $\tilde{\Delta}|S|^2_{\tilde{g}}$

use connection $\tilde{\nabla}$ for $\tilde{g}$

repeated indices are contracted using $\tilde{g}$

e.g. $$|S|^2_{\tilde{g}} = \tilde{g}^{jk}\,\tilde{g}^{a\bar{b}}\,\tilde{g}_{p\bar{q}}\,S^{p}_{ja}\,\overline{S^{q}_{kb}}$$

$$= S^{p}_{ja}\,\overline{S^{p}_{ja}}$$

$$\tilde{\Delta}|S|^2_{\tilde{g}} = \tilde{\nabla}_p\tilde{\nabla}_{\bar{p}}\left(S^i_{jk}\,\overline{S^i_{jk}}\right)$$

$$= \left(\tilde{\nabla}_p\tilde{\nabla}_{\bar{p}}S^i_{jk}\right)\overline{S^i_{jk}} + S^i_{jk}\,\overline{\tilde{\nabla}_{\bar{p}}\tilde{\nabla}_p S^i_{jk}}$$

$$+ \tilde{\nabla}_p S^i_{jk}\,\overline{\tilde{\nabla}_p S^i_{jk}} + \tilde{\nabla}_{\bar{p}} S^i_{jk}\,\overline{\tilde{\nabla}_{\bar{p}} S^i_{jk}}$$

$$\geq \left(\tilde{\nabla}_p\tilde{\nabla}_{\bar{p}}S^i_{jk}\right)\overline{S^i_{jk}} + S^i_{jk}\,\overline{\tilde{\nabla}_{\bar{p}}\tilde{\nabla}_p S^i_{jk}}$$

commuting derivatives:

$$\tilde{\nabla}_{\bar{p}}\tilde{\nabla}_p S^i_{jk} = \tilde{\nabla}_p\tilde{\nabla}_{\bar{p}}S^i_{jk} + \tilde{R}_j{}^m{}_{p\bar{p}}\,S^i_{mk} + \tilde{R}_k{}^m{}_{p\bar{p}}\,S^i_{jm}$$

$$- \tilde{R}_m{}^i{}_{p\bar{p}}\,S^m_{jk}$$

$$= \tilde{\nabla}_p\tilde{\nabla}_{\bar{p}}S^i_{jk} + \tilde{R}_j{}^m S^i_{mk} + \tilde{R}_k{}^m S^i_{jm} - \tilde{R}_m{}^i S^m_{jk}$$

From the equation $\det(\tilde{g}) = e^F \det(g)$

$$\tilde{R}_{j\bar{k}} = R_{j\bar{k}} - \partial_j \partial_{\bar{k}} F$$

$$\Rightarrow |\tilde{R}_{j\bar{k}}|_{\tilde{g}} < C \quad (\text{dep. on } (M,g), F)$$

$$\Rightarrow |\tilde{\nabla}_{\bar{p}} \tilde{\nabla}_p S^i_{jk}|_{\tilde{g}} \leq |\tilde{\nabla}_p \tilde{\nabla}_{\bar{p}} S^i_{jk}|_{\tilde{g}} + C_1 |S|_{\tilde{g}}$$

$$\tilde{\nabla}_p \tilde{\nabla}_{\bar{p}} S^i_{jk} = \tilde{\nabla}_p \partial_{\bar{p}} (\tilde{\Gamma}^i_{jk} - \Gamma^i_{jk})$$

$$= -\tilde{\nabla}_p (\tilde{R}_j{}^i{}_{k\bar{p}} - R_j{}^i{}_{k\bar{p}})$$

Bianchi $\Rightarrow$ $\tilde{\nabla}_p \tilde{R}_j{}^i{}_{k\bar{p}} = \tilde{\nabla}_k \tilde{R}_j{}^i{}_{p\bar{p}} = \tilde{\nabla}_k \tilde{R}_j{}^i$

$$= \nabla_k \tilde{R}_j{}^i + (\tilde{\nabla}_k - \nabla_k) \tilde{R}_j{}^i$$

So $|\tilde{\nabla}_p \tilde{R}_j{}^i{}_{k\bar{p}}|_{\tilde{g}} \leq C + C|S|_{\tilde{g}}$

$$\rightsquigarrow |\tilde{\nabla}_p \tilde{\nabla}_{\bar{p}} S^i_{jk}|_{\tilde{g}} \leq C + C|S|_{\tilde{g}}$$

Combining estimates:

$$\tilde{\Delta} |S|^2_{\tilde{g}} \geq -(C + C|S|_{\tilde{g}}) |S|_{\tilde{g}}$$

$$\geq -C_2 |S|^2_{\tilde{g}} - C_3 \qquad (2ab \leq a^2 + b^2)$$

$\square$

<u>Conclusion</u> We have $\|f\|_{C^0}, \|\partial_i \partial_j \partial_{\bar{k}} f\|_{C^0} < C$

$$\Rightarrow \|\Delta f\|_{C^1} < C$$

Schauder $\Rightarrow$ $\|f\|_{C^{2,\alpha}} \leq C(\|\Delta f\|_{C^\alpha} + \|f\|_{C^0}) \leq C'$

C' depends on (M,g) and $\|F\|_{C^3}$

Recall continuity method

$$\left.\begin{aligned} \det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f) &= e^{tF + c_t} \det(g_{j\bar{k}}) \\ g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f &> 0 \\ \sup_M f &= 0 \end{aligned}\right\} (*_t)$$

Let $S = \{ t \in [0,1] \mid (*_t) \text{ has solution} \}$

Proof that S is closed: Let $t_i \in S$, $t_i \to t_\infty$

$$\det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_{t_i}) = e^{t_i F + c_{t_i}} \det(g_{j\bar{k}})$$
$$= e^{F_i} \det(g_{j\bar{k}})$$

to bound $\|F_i\|_{C^3}$ still need to bound c_{t_i}.

If $f_{t_i}(p) = \sup_M f_{t_i}$, then

$$\det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_{t_i})(p) \leq \det(g_{j\bar{k}})(p)$$

so $e^{t_i F(p) + c_{t_i}} \leq 1$

$$\Rightarrow \quad c_{t_i} \leq -t_i F(p) < C$$

similarly $c_{t_i} > -C$.

$\Rightarrow \quad \|F_i\|_{C^3} < C$ indep. of i

A priori estimates $\Rightarrow \|f_{t_i}\|_{C^{2,\alpha}} < C$

Arzela-Ascoli $\Rightarrow$ subsequence of $f_{t_i} \to f_\infty$ in $C^{2,\beta}$, $\beta < \alpha$

$c_{t_i} \to c_\infty$

Pass to limiting eq^n

$$\det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_\infty) = e^{t_\infty F + C_\infty} \det(g_{j\bar{k}})$$

so $(*_{t_\infty})$ has a solution, i.e. $t_\infty \in S$

and S is closed. $\Rightarrow 1 \in S$.

<u>Last step</u>: uniqueness (HW)

Completes the proof of

<u>Thm</u> (Yau) Let (M, ω) be compact Kähler,

$\alpha \in 2\pi c_1(M)$ real $(1,1)$-form,

then $\exists$ unique $\eta \in [\omega]$ s.t. $\mathrm{Ric}(\eta) = \alpha$.

<u>Negative KE metrics</u>

<u>Thm</u> (Yau) Let (M, ω) be compact Kähler,

$\omega \in -2\pi c_1(M)$. Then $\exists$ unique $\eta \in [\omega]$

s.t. $\underset{\substack{\Uparrow \\ 2\pi c_1(M)}}{\mathrm{Ric}(\eta)} = -\eta$

<u>Proof</u> Write as PDE:

Let F be defined by

$$\mathrm{Ric}(\omega) + \omega = \sqrt{-1}\,\partial\bar{\partial} F$$

Suppose $f : M \to \mathbb{R}$ satisfies

$$\det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f\right) = e^{F+f} \det(g_{j\bar{k}}) \qquad (*)$$

(g is metric with Kähler form ω), $\eta_{j\bar{k}} = g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f$

$$\Rightarrow \quad \mathrm{Ric}(\eta) = -\sqrt{-1}\,\partial\bar{\partial} F - \sqrt{-1}\,\partial\bar{\partial} f + \mathrm{Ric}(\omega)$$
$$= -\omega - \sqrt{-1}\,\partial\bar{\partial} f = -\eta$$

So $\mathrm{Ric}(\eta) = -\eta$

Continuity method can be used to solve $(*)$.

<u>Positive KE metrics</u>

PDE : $\det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f\right) = e^{F-f} \det(g_{j\bar{k}})$